

A Complete Proof of Conjecture TLMC-00000001206: the Nim-Sum Period Law

Solution submission

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Abstract

We prove Conjecture TLMC-00000001206 of the Last Math Competition, which asserts that the period of the pointwise nim-sum of two periodic Grundy sequences divides the least common multiple of the two periods. We give an elementary, self-contained, unconditional proof, together with a machine-verified Lean 4 formalization (file `NimSumPeriod.lean`, no `sorry`). In fact we prove a stronger statement: the conclusion holds for *arbitrary* $\mathbb{N}$ -valued periodic sequences under *any* pointwise binary operation; the conjecture for Grundy sequences and the nim-sum follows a fortiori.

1 The conjecture

The conjecture as recorded in `conjectures/00000001206.md` reads:

Conjecture 1.1 (TLMC-00000001206; nim-sum period). (*English*) Nim-sum closure of Sprague–Grundy functions: the period of the pointwise nim-sum of two periodic Grundy sequences divides the lcm of the two periods.

(*Chinese*) Sprague–Grundy nim- Grundy nim- lcm

2 Definitions and notation

Throughout, all sequences are functions $g: \mathbb{N} \rightarrow \mathbb{N}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$.

Definition 2.1 (Period). A positive integer $p \geq 1$ is a *period* of a sequence $g: \mathbb{N} \rightarrow \mathbb{N}$ if

$$g(n + p) = g(n) \quad \text{for all } n \in \mathbb{N}.$$

A sequence is *periodic* if it has at least one period.

Definition 2.2 (Nim-sum). The *nim-sum* (bitwise exclusive or) of two nonnegative integers is written $x \oplus y$; explicitly, if $x = \sum_i x_i 2^i$ and $y = \sum_i y_i 2^i$ with digits $x_i, y_i \in \{0, 1\}$, then $x \oplus y = \sum_i (x_i + y_i \bmod 2) 2^i$. The *pointwise nim-sum* of two sequences g_1, g_2 is the sequence $n \mapsto g_1(n) \oplus g_2(n)$.

Definition 2.3 (Minimal period). The *minimal period* (or *the period*) $\pi(g)$ of a periodic sequence g is the least positive period:

$$\pi(g) = \min \{p \geq 1 : p \text{ is a period of } g\}.$$

The minimum exists because the set is a nonempty set of positive integers.

Definition 2.4 (Grundy sequence). A *Grundy sequence* is the Sprague–Grundy function g of an impartial combinatorial game (with positions indexed by $\mathbb{N}$): $g(n) = \text{mex}\{g(m) : m \in \text{opt}(n)\}$, where mex is the least excluded nonnegative integer. In particular, every Grundy sequence is an $\mathbb{N}$ -valued sequence.

3 The theorem and its complete proof

Theorem 3.1 (Nim-sum period law). *Let $g_1, g_2 : \mathbb{N} \rightarrow \mathbb{N}$ be periodic sequences with periods $p_1, p_2 \geq 1$, respectively. Then the pointwise nim-sum $g = (n \mapsto g_1(n) \oplus g_2(n))$ is periodic, and its minimal period divides $\text{lcm}(p_1, p_2)$:*

$$\pi(g) \mid \text{lcm}(p_1, p_2).$$

Proof. The proof uses only the fact that the nim-sum is a *function* of the two values $g_1(n)$ and $g_2(n)$; we isolate three elementary steps.

Step 1 (multiples of a period are periods). Let f be a sequence with a period $p \geq 1$. We show by induction on k that $k \cdot p$ is a period of f for every $k \in \mathbb{N}$. For $k = 0$ this says $f(n+0) = f(n)$, which is trivial. Assume $k \cdot p$ is a period. Then for every n ,

$$f(n + (k+1)p) = f((n + kp) + p) \stackrel{(*)}{=} f(n + kp) = f(n),$$

where $(*)$ holds because p is a period (applied at the index $n + kp$), and the last step is the induction hypothesis. Hence $(k+1) \cdot p$ is a period, completing the induction.

Step 2 (the lcm is a period of the nim-sum). Write $L = \text{lcm}(p_1, p_2)$. Since $p_1 \mid L$, there is $a \in \mathbb{N}$ with $L = a p_1$; by Step 1 (with $k = a$), $g_1(n + L) = g_1(n)$ for all n . Similarly $p_2 \mid L$ gives $b \in \mathbb{N}$ with $L = b p_2$ and $g_2(n + L) = g_2(n)$ for all n . Consequently, for every n ,

$$g(n + L) = g_1(n + L) \oplus g_2(n + L) = g_1(n) \oplus g_2(n) = g(n),$$

so L is a period of g . In particular g is periodic, and by Definition 2.3 its minimal period $\pi = \pi(g)$ exists and satisfies $1 \leq \pi \leq L$.

Step 3 (the minimal period divides every period). Let $q \geq 1$ be any period of f , and let π be the minimal period of f ; we claim $\pi \mid q$. Perform Euclidean division: $q = c\pi + r$ with $c \in \mathbb{N}$ and $0 \leq r < \pi$. By Step 1, $c\pi$ is a period of f , so for every n ,

$$f(n + r + c\pi) = f(n + r).$$

On the other hand $n + r + c\pi = n + q$, and since q is a period, $f(n + q) = f(n)$. Combining,

$$f(n + r) = f(n + q) = f(n) \quad \text{for all } n.$$

Thus r is a period of f . If $r \geq 1$, then r is a positive period with $r < \pi$, contradicting the minimality of π . Hence $r = 0$, i.e. $\pi \mid q$, proving the claim.

Conclusion. Apply Step 3 to $f = g$, $\pi = \pi(g)$, and the period $q = L$ supplied by Step 2: $\pi(g) \mid L = \text{lcm}(p_1, p_2)$, which is the assertion of the theorem. $\square$

4 Derivation of Conjecture 1.1

Corollary 4.1. *Conjecture 1.1 holds.*

Proof. Let g_1, g_2 be periodic Grundy sequences with (minimal) periods p_1, p_2 . Grundy sequences are $\mathbb{N}$ -valued (Definition 2.4), and the pointwise nim-sum of two $\mathbb{N}$ -valued sequences is $\mathbb{N}$ -valued. Theorem 3.1 applies to arbitrary $\mathbb{N}$ -valued periodic sequences; hence the minimal period of $n \mapsto g_1(n) \oplus g_2(n)$ divides $\text{lcm}(p_1, p_2)$, which is precisely the assertion of Conjecture 1.1. $\square$

Remark (Strength of the result). The proof of Theorem 3.1 uses nothing about the nim-sum beyond its being a well-defined binary operation on the value set: Steps 1–3 are statements about periods alone. The identical argument proves: *for any binary operation $\star: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ (pointwise addition, multiplication, maximum, bitwise and/or/xor, ...), the minimal period of $n \mapsto g_1(n) \star g_2(n)$ divides $\text{lcm}(p_1, p_2)$.* The nim-sum case of the conjecture is one instance. The Lean formalization is carried out for a completely arbitrary binary operation, making this generality machine-checked.

Remark (Sharpness). Both divisibility conclusions are optimal in general: taking g_2 constant, the nim-sum $g_1 \oplus g_2$ need not have any period smaller than the lcm of the individual minimal periods, so no strengthening of the divisibility (e.g. to a fixed proper factor of $\text{lcm}(p_1, p_2)$) is possible without further hypotheses.

5 Lean 4 formalization

The file `NimSumPeriod.lean` (toolchain `leanprover/lean4:v4.31.0`, Mathlib `v4.31.0`) contains a complete machine-checked proof:

- `IsPeriod f p` — Definition 2.1;
- `minPeriod f` — Definition 2.3, via least positive period;
- `IsPeriod.mul, minPeriod.dvd` — Steps 1 and 3;
- `pointwise_period` — Step 2 and Theorem 3.1, proved for an *arbitrary* binary operation `op` in place of `Nat.xor`;
- `nimsum_period` — Theorem 3.1 specialized to the nim-sum ($\hat{\hat{}}) = \text{Nat.xor}$;
- `conjecture1206_true` — the faithful statement of Conjecture 1.1 for $\mathbb{N}$ -valued sequences, proved.

The file compiles with no errors and contains no `sorry`.

6 Conclusion

Conjecture TLMC-00000001206 is **proved**: for any two $\mathbb{N}$ -valued periodic sequences with periods $p_1, p_2 \geq 1$, the minimal period of their pointwise nim-sum divides $\text{lcm}(p_1, p_2)$ (Theorem 3.1, Corollary 4.1). The proof is elementary, unconditional, and fully formalized in Lean 4.